Zihao Jin

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Shenzhen, Guangdong, China | Research: Multimodal LLMs, Computer Vision,Robotics

Education

Sun Yat-sen University
Guangdong, China

School of Artificial Intelligence
Sep. 2023 – Expected Jun. 2027

- GPA: 86/100. Coursework: Computer Vision (97), Machine Learning (96), Human-Computer Interaction (94), Numerical Computing (96), Digital Signal Processing (94), Game Theory (90).
- Honors: SYSU Outstanding Student Scholarship; Provincial Mathematical Modeling Award; MCM Honorable Mention; 3rd Place, National Collegiate Dragon Boat Championship; Excellence Award, "Win at SYSU" Competition.

Publications & Research Experience

Nereus: A Large-Scale Underwater Dataset for Fine-Grained Attribute Understanding and Grounded Counting
Nov. 2025 – May 2026

Publication, First-Author | NeurIPS 2026 accepted

- Built and cleaned an underwater multimodal dataset with 88K images, 2.68M QA pairs, 136K objects, and 65K point annotations; helped design a 136-dimensional object-level attribute taxonomy.
- Constructed region-aware counting, camouflage-aware counting, and density-hotspot reasoning tasks from point annotations; evaluated Qwen, InternVL, LLaVA, and related models for benchmark analysis.

RSI-Harness: Recursive Skill Composition and Acquisition for Self-Improving Embodied Agents
Jul. 2026 – Present

Research Project, Embodied AI and Humanoid Robot Learning

- Designed RSI-Harness, a hierarchical recursive self-improvement framework that transforms human demonstrations and Real2Sim experience into reusable robot skills; introduced failure attribution to distinguish skill-composition errors from missing or insufficient capabilities.
- Architected a Skill Harness with a planner/composer, executor, Critic/Challenger, verification, and persistent memory; Outer RSI replans skill programs, while Inner RSI generates targeted data to adapt, verify, and store new policies.

Ego2Robot: Egocentric Data Collection and Imitation Learning Pipeline for Unitree G1
May 2026 – Present

Research Project, Humanoid Robot Data Pipeline

- Built a Unitree G1 teleoperation pipeline with RealSense D435i, PICO/NoMachine, and GR00T-WholeBodyControl, enabling synchronized image streaming and manipulation data collection.
- Established a real-robot imitation learning workflow for cup-grasping tasks, covering data cleaning, LeRobot/GR00T format alignment, fine-tuning preparation, and evaluation.

Skills

Languages & Tools: Python, C++, MATLAB, Linux, Git, Gradio
ML / Vision: PyTorch, OpenCV, multimodal LLMs/VLMs, Diffusion Policy/EDP, object detection, pose estimation
Robotics / Data: Isaac Lab, GR00T, LeRobot, RealSense, ZMQ, data collection, benchmark construction, model evaluation

Additional Information

- Languages: Mandarin Chinese (native); English: IELTS 7.0 (6.5) / CET-4 / CET-6.
- Research strengths: multimodal dataset construction, embodied robot data collection, model evaluation, and cross-disciplinary application exploration.